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ORIGINAL INVESTIGATION

The Validity of Perceived Recovery Status as a Marker of Daily Recovery Following a High-Volume Back-Squat Protocol

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Although a variety of tools to monitor recovery have been developed, many are impractical for daily use due to cost, time, and challenges with interpretation. The Perceived Recovery Status (PRS) scale was recently developed as an expeditious, noninvasive tool to assess recovery status. While PRS has been strongly associated with repeated sprinting performance, a paucity of research exists relating PRS and performance recovery following resistance exercise. ***Purpose:*** The purpose of this study was to evaluate the sensitivity of PRS as a subjective marker of recovery up to 72 hours after a high-volume back-squat protocol. ***Methods:*** Eleven resistance-trained men reported to the laboratory on 5 separate occasions (1 familiarization session and 4 testing sessions). The first testing session was considered the baseline session and consisted of a nonfatiguing performance assessment (ie, countermovement jumps and back squats) and a fatiguing back-squat protocol of 8 sets of 10 at 70% 1-repetition maximum separated by 2 minutes of recovery. Participants returned 24, 48, and 72 hours following baseline to provide a PRS rating and complete the performance assessment. ***Results:*** Repeated-measures correlations revealed strong associations between PRS countermovement jump ($r = .84$) and mean bar velocity ($r = .80$) (both $P < .001$). ***Conclusions:*** The current findings suggest that PRS can be used as a method to effectively assess daily recovery following a fatiguing bout of resistance exercise. Practitioners are cautioned that the relationship between PRS and performance recovery is individualized, and equivalent PRS scores between individuals are not indicative of similar recovery.

Keywords: perception, resistance training, monitoring

The goal of any resistance-training program is to optimize athletic performance via adaptation by eliciting a stress response through the manipulation of training variables (eg, frequency, load, volume). While the magnitude of the stress response is an important component to drive adaptation, equally important is the recovery from said stressor. Indeed, acute and chronic imbalances between these factors can have deleterious physiological, psychological, and performance-related effects.[1,2] Monitoring acute and chronic recovery is, therefore, of upmost importance for maximizing resistance training adaptations.

Recovery is a complex construct potentially involving the assessment of numerous physiological and psychological indicators.[1–3] As such, an array of instruments and techniques have been created to accurately and reliably assess various aspects of recovery.[4,5] In laboratory settings, biomarkers of stress and muscle damage are often used to assess physiological strain. However, these methods are impractical for daily monitoring due to limited accessibility to specialized equipment, personnel for collecting and analyzing samples, and like any assessment tool, high intersubject and intrasubject variability.[6]

In field settings, recovery is often assessed by comparing current performance indicators (eg, one-repetition maximum [1RM], bar velocity metrics, vertical jump height, etc) to maximal performance values obtained previously from a baseline or initial testing session completed when the individual was fully recovered. Performance testing, however, requires time and fatigue inducing maximal or near-maximal effort, making it an inappropriate assessment tool for athletes who experience periods of exhaustion during their competitive season.[7] Multi-item psychological questionnaires have been developed to subjectively quantify stress and recovery from exercise in both laboratory and field settings when objective performances tests are unavailable or inappropriate.[1,8] Careful consideration must be given to the length of the questionnaire and the nature of the questions as longer time requirements to complete a survey can be burdensome, in turn affecting compliance and data quality.[9] Collectively, the currently available methods for assessing recovery status and readiness to perform may not be widely available, practical, or time efficient. Therefore, the need for ecologically valid field-based methods that gauge recovery status warrants further research.

The Perceived Recovery Status (PRS) scale was developed as a practical way for individuals to subjectively gauge recovery and estimate subsequent performance.[3] The PRS scale is a single-item, 0- to 10-point scale, with verbal anchors defining the numerical indicators of both recovery and expected performance. A PRS score of zero indicates poor recovery and potentially poor performance, whereas a score of 10 indicates optimal recovery and performance. Despite the limited empirical research exploring PRS, the studies that have investigated this topic report encouraging results. For example, Laurent et al[3] found that PRS was strongly correlated with changes in sprint performance 24, 48, and 72 hours after a repeated sprinting protocol in sprint-trained individuals ($r = -.63$), indicating that lower PRS scores were associated with longer time to completion (poorer performance). Korak et al[10] assessed recovery using both PRS and objective performance indicators (repetitions performed to failure) at 24 and

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48 hours following a bout of whole-body resistance exercise. Researchers observed that individuals were better able to mirror their baseline performance at 48 hours compared with 24 hours of recovery, which also coincided with higher PRS (ie, greater perceived recovery). While Korak et al[10] demonstrated that PRS can dichotomize recovery status into recovered and unrecovered, research has yet to establish if PRS correlates with performance indicators measured at multiple time points across the recovery continuum that it is known following resistance exercise.[1,2] Therefore, the purpose of this study was to evaluate the sensitivity of PRS as a subjective marker of recovery by directly quantifying its relationship with objective indicators of performance at multiple time points (24, 48, and 72 h) following a high-volume back squat protocol.

Methods

Subjects

Eleven apparently healthy, resistance-trained men (age = 22.9 [3.4] y, height = 179.9 [7.8] cm, weight = 89.9 [10.7] kg, body fat = 13.7% [4.3%], back squat 1RM = 156.5 [29.4] kg) participated in this study. An a priori power analysis, using G*Power (version 3.9.1.4; University of Dusseldorf, Dusseldorf, Germany), indicated a minimum of 10 participants, with 4 repeated measures, were needed to yield a power of 0.80 for detecting a correlation .50 with significance set at $\alpha = .05$. To be included in the study, participants must have been resistance training for >1 year (all participants reported 5+ y of consistent resistance training experience), be asymptomatic of cardiovascular, metabolic, and renal disease, report no musculoskeletal injuries, and been a nonsmoker for at least 6 months. Participants refrained from alcohol, stimulants, and nonprescription drugs 12 hour prior to participating in the study and intense physical activity 72 hours prior to and throughout the entirety of the study. Adherence to these guidelines was assessed by a 24-hour history questionnaire completed at the beginning of each visit. Written informed consent was obtained prior to participation, and the study was approved by the University of Alabama institutional review board.

Experimental Design

Participants visited the laboratory on 5 occasions (1 familiarization and 4 testing sessions) in this repeated measure design (Figure 1), where they completed a nonfatiguing recovery assessment at all 4 testing sessions (baseline, 24, 48, and 72 h) and a high-volume back squat protocol during the baseline testing session. Testing sessions were completed within 1 hour of waking, with all testing sessions occurring at the same time each day (±30 min) to control for diurnal variation in performance.

Familiarization Session

The first visit to the laboratory consisted of paperwork (eg, informed consent, health history, and 24-h recall questionnaire), obtaining anthropometric measurements, establishing the 1RM for back squat, and familiarization to the performance protocol. Standing height was recorded to the nearest 0.1 cm via manual stadiometer (Seca 213; Seca Ltd, Hamburg, Germany); and nude body mass was recorded to the nearest 0.1 kg using a digital scale (Tanita BWB-800; Tanita Corp, Tokyo, Japan). Body fat percentage was estimated using a 3-site skinfold technique.[11] Participants then completed a dynamic warm-up consisting of 5-minute cycling at a self-selected intensity and 10 repetitions for each of the following: bodyweight squats, walking lunges, and dynamic hamstring and quadriceps stretches. A standardized 1RM protocol was used to assess back squat 1RM.[12] Participants completed 3 warm-up sets of back squats consisting of 5 to 10 repetitions at approximately 50% 1RM, 3 repetitions at approximately 85% 1RM, and 1 repetition at approximately 90% 1RM. Determination of 1RM was established using incremental

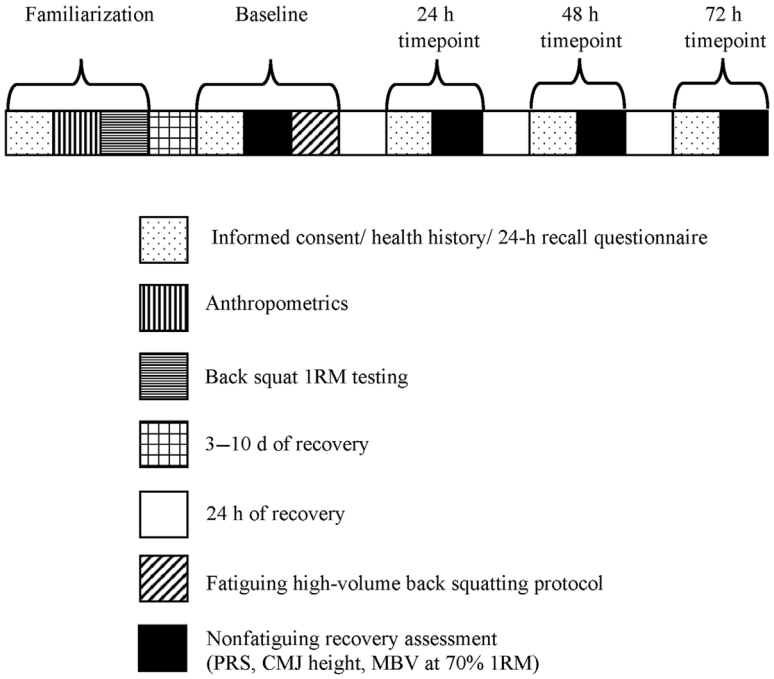

Figure 1 — Schematic illustration of study protocol. CMJ indicates countermovement jump; MBV, mean bar velocity; PRS, perceived recovery status; 1RM, 1-repetition maximum.

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weight increases. Following a successful lift, participants were given 3 to 5 minutes of recovery before completing the next attempt where load was increased by 10% to 20%. If an attempt was unsuccessful then the load was dropped by 5% to 15%, and participants attempted the weight following 3 to 5 minutes of recovery. This protocol was continued until failure, which occurred within 5 attempts for all study participants.

Testing Sessions

Participants first completed a 24-hour recall questionnaire before the dynamic warm-up protocol that was performed during the familiarization session. A member of the research team then presented the PRS scale to the participants and gave a thorough explanation of the scale before participants provided a subjective rating of their recovery.

Following the subjective assessment of recovery, participants performed 3 countermovement jumps (CMJs) with 1 minute of recovery between each jump. Participants stood on a pair of portable force plates (Kistler 9286BA 10kN; Kistler Group, Winterthur, Switzerland) with their hands anchored to their hips and performed a maximal vertical jump. Jump height was calculated using peak take-off velocity (Measurement, Analysis & Reporting Software; Kistler Group). The mean of 3 attempts was calculated and served as their baseline measure in subsequent analyses.

Participants then completed 2 × 1 back squat at 70% 1RM with 3 minutes of recovery between attempts using maximal effort. Mean bar velocity (MBV) was recorded using linear force transducer calibrated using manufacturer guidelines (GymAware; Kinetic Performance Technology Pty., Ltd, Mitchell, ACT). The mean of 2 attempts was calculated and served as their baseline measure for analyses.

Following the baseline nonfatiguing recovery assessment, participants completed a high-volume back squat protocol consisting of 8 sets of 10 repetitions at 70% 1RM with 2-minute recovery between each set. For a repetition to be deemed successful, the participants' thighs must be parallel to the floor during the eccentric portion of the lift with their hips and knees being fully extended during the concentric portion of the lift. This was explained to each participant prior to the start of protocol. To confirm a successful attempt, one researcher was positioned beside the participant to ensure intended eccentric and concentric positions were reached. Failure to complete 10 successful repetitions resulted in a 30-second rest before participants completing the remaining repetitions in that set. Three attempts were given to complete the remaining repetitions for a given set; in the event they failed all 3 attempts, the set concluded, a 2-minute recovery timer was started, and the load was decreased by 10% before starting the next set. Five out of the 11 participants required a reduction in the weight lifted during the protocol, ranging between 1 and 4 reductions per individual. Repetitions that were not successfully completed within the 8 sets were added to an additional set at the end to ensure 80 total repetitions were completed. Participants returned to the laboratory 24, 48, and 72 hours after the baseline trial where they provided a subjective recovery rating using the PRS scale and completed the nonfatiguing recovery assessment (Figure 1).

Statistical Analysis

Recovery scores were calculated for CMJ and MBV to standardize the magnitude of fatigue for all measures across all time points. Recovery scores were calculated by dividing the raw performance metric at the time point of interest by the raw baseline performance metric and multiplying by 100. Thus, recovery scores can be interpreted as a percentage of an individual's baseline performance (percentage recovered). In addition, a daily recovery score was calculated for the nonfatiguing performance tests by averaging recovery scores for MBV and CMJ within a session. Repeated-measures correlations were used to assess the intraindividual relationship between PRS and performance in the raw and recovery scores achieved for the nonfatiguing recovery assessments across baseline, 24, 48, and 72 hours recovery time points. This method allows for the modeling of a common association between 2 measures over time, while adjusting for differences between individual participants. Bootstrapped, 95% confidence intervals (CIs) were also constructed to provide a level of precision for the estimated correlation coefficient. All data were analyzed using R (version 3.4.0.1; R Core Team, Vienna, Austria) using the *rmcorr* package.[13] Statistical significance was determined a priori at $\alpha \leq .05$. Data are presented as mean (SD) unless otherwise stated.

Results

Daily average and participant-level trends for PRS and performance metrics are outlined in Table 1 and Figure 2, respectively. Repeated-measures correlations revealed PRS to be strongly correlated to both raw ($r = .844$; 95% CI, .73–.913; $P < .001$) and

Table 1 Average Perceived and Performance Measures at Baseline and Over 72 Hours of Recovery Following a Fatiguing High-Volume Back-Squat Protocol (N = 11)

	Baseline	24 h	48 h	72 h
Perceived Recovery Status scale[a]	8.3 (1.3)	4.6 (2.0)	4.8 (1.8)	6.2 (2.2)
Countermovement jump				
Height, m	0.39 (0.05)	0.33 (0.05)	0.33 (0.07)	0.33 (0.09)
Recovery, %	100 (0)	84 (6)	84 (14)	85 (19)
Mean bar velocity				
Velocity, m/s	0.67 (0.07)	0.55 (0.14)	0.59 (0.13)	0.59 (0.14)
Recovery, %	100 (0)	81 (16)	87 (15)	87 (15)

Note: The Perceived Recovery Status scale (Laurent et al[3]) is a single-item, 0- to 10-point scale with 2 sets of verbal anchors defining the numerical indicators of both recovery and expected performance, where 0 = *very poorly recovered* (poor performance) and 10 = *fully recovered* (optimal performance).

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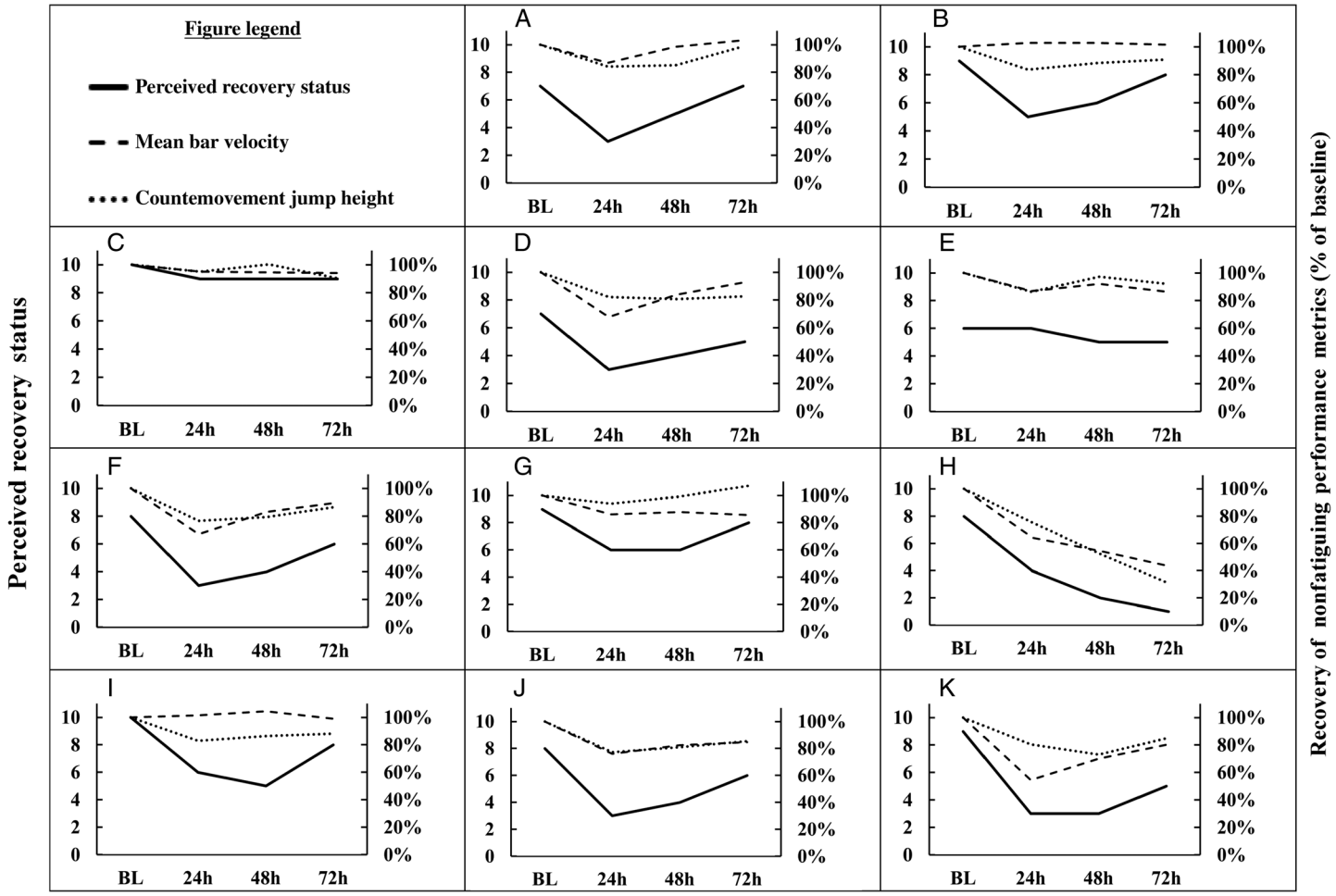

Figure 2 — Participant-level plots (A–K) of perceived recovery status and nonfatiguing performance tests at baseline and over 72 hours of recovery following a high-volume back-squat protocol. Each letter indicates a different participant.

recovery of CMJ height ($r = .843$; 95% CI, .743–.913; $P < .001$), as displayed in Figure 3. In addition, PRS shared a strong, positive relationship with raw ($r = .800$; 95% CI, .631–.899; $P < .001$) and recovery of MBV ($r = .805$; 95% CI, .621–.904; $P < .001$; Figure 3). Intraindividual differences between performance assessments and PRS ratings can be observed in Table 2.

Discussion

Successful training programs drive adaptation through the manipulation of variables to elicit a stress response. While much of the emphasis has been placed upon enhancing this stress response, fully optimized training must balance stress with adequate recovery. Recovery markers such as counter movement jump, MBV, and creatine kinase have been proposed to effectively assess recovery efficacy, but they are often impractical for daily use in an applied setting. Thus, the purpose of this study was to determine the validity of the PRS scale as a marker of recovery of performance following a bout of high-volume, fatiguing resistance exercise. The major finding from the current study was that individuals could accurately relate their subjective recovery status via PRS to objective markers of performance over 72 hours of recovery following a fatiguing high-volume back squat protocol.

Recovery is a complex construct involving a myriad of physiological and psychological indicators, with numerous metrics developed to assess individual recovery status. The current study defined recovery from a practical perspective, where recovery status would manifest itself as decrements in performance.[7] Performance can also be defined in a variety of ways, and as such, multiple assessments of neuromuscular performance were used to thoroughly evaluate recovery of performance. CMJ height and MBV were chosen based upon high degrees of reliability[14] and their direct[15] or indirect relationship[16,17] with neuromuscular performance and fatigue outcomes. For example, various studies have used the inverse relationship between bar velocity and relative load to predict maximal strength. Therefore, decrements in MBV are likely the manifestation of decreases in maximal strength due to insufficient recovery.

Similar trends were observed for both CMJ and MBV; that is, a considerable decrease 24 hour following the back squat protocol that remained suppressed through the 72-hour follow-up (Table 1). This is not surprising, as previous research has shown that higher volume exercise has deleterious effects on both bar velocity and CMJ performance 48- to 72-hour postexercise.[4,18] While average performance remained suppressed throughout the study, there was a large amount of heterogeneity in both magnitude and pattern of recovery throughout 72 hours. For example, one participant (Figure 2H) saw a continuous decline in MBV and CMJ performance (approximately 40% of baseline) while another participant (Figure 2C) was almost fully recovered (approximately 95% of

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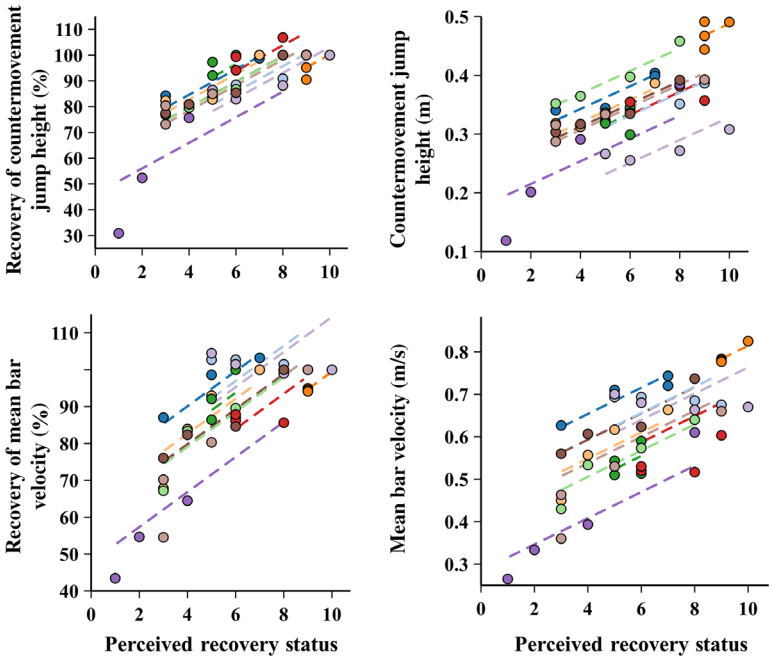

Figure 3 — Scatterplots depicting the relationship between perceived recovery status and raw and recovery scores for performance testing. Note: Dotted lines are presented to demonstrate the subject-level differences in *y*-intercept values modeled in the analysis. Line length depicts the range of perceived recovery status reported by each subject. See online color version of this article for lines and points color-coded by subject.

Table 2 Frequency Distribution of PRS Scores in Relation to Mean Recovery Score (N = 11)

	Mean recovery score[a]							
PRS score	<65	65%–70%	70%–75%	75%–80%	80%–85%	85%–90%	90%–95%	95%–100%
1	1							
2	1							
3		1	2	2		1		
4			1		3			
5					1	2	3	1
6					1	2	3	2
7								3
8							1	5
9							1	5
10								2

Abbreviation: PRS, Perceived Recovery Status scale.
[a]Mean recovery score was calculated as the average recovery score of mean bar velocity and countermovement-jump recovery scores at each time point.

baseline) at 24 hours. The fact that such large individual differences were observed in our sample despite similar training ages provides further evidence that monitoring individualized recovery is a critical component of a successful training program.

The PRS decreased substantially from baseline to 24 hours and remained dampened at 48 hours following the fatiguing protocol, signifying that PRS is sensitive to perturbations to homeostasis 24 hours following a stressor. This is not surprising and, indeed, expected as similar trends have been observed in both resistance exercise[10,19] and repeated sprint work.[20,21] Korak et al[10] observed significant decreases in PRS 24 hours following a whole-body resistance bout, though PRS significantly improved at 48 hours while Sikorski et al[19] found greater overall decrements in PRS, which were still markedly lower at 48 hours as compared with Korak et al.[10] These incongruent findings can be explained by the resistance protocols as the higher numbers of total repetitions and similar (*if not slightly higher*) relative workloads in the Sikorski et al[19] study could have caused greater overall metabolic and mechanical stress, which would then be reflected in a lower recovery.[22,23] Interestingly, PRS in the current study was lower than reported by Korak et al[10] despite total repetitions being higher in the latter. It should be noted that although total volume was lower in the current study, that lower volume was concentrated on one exercise as compared with 10 exercises. This *could* cause a greater localized stress responses and nociceptive signaling resulting in greater decrements in PRS observed in the current study. Collectively, this suggests that PRS may be mediated by whole-body and individual muscle afferent feedback.

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Although this is the first study to directly quantify the intraindividual relationship between PRS and objective performance indicators following resistance exercise, Laurent et al[3] reported similar findings between PRS and recovery from a repeated sprinting protocol at 24, 48, and 72 hours. In regards to resistance exercise, Sikorski et al[19] reported PRS to be strongly correlated with creatine kinase and muscular soreness 48 hours following a multiexercise resistance bout. Though not measured, the authors alluded to a possible relationship between PRS and objective performance following resistance exercise.[19] While plasma creatine kinase concentration is the product of release and clearance and may not be a direct assessment of muscle damage, numerous studies have found elevated creatine kinase levels following a bout of high-volume resistance exercise to be associated with decreases in performance.[4,24] Therefore, the results of the current study support previous findings of high correlations between PRS and performance decrements.

While there is an observable trend of increasing PRS values coinciding with performance, the variability of performance recovery given a PRS score is clear (Table 2). For example, a PRS score of 6 was reported 8 times throughout the study, which coincided with objective recovery scores ranging from 84% to 100%. To look at it from an objective standpoint, there were a total of 19 time points in which participants produced recovery scores of 95% to 100%. Within this same recovery range, PRS scores varied from 5 to 10. While this appears to call in to question the validity of PRS, there are additional factors to consider. Primarily, objective recovery was calculated based on performance relative to baseline with assumption that refraining from resistance training at least 72 hours before testing would permit physiological recovery and maximize performance. However, while participants were given general guidelines to control for both physiological and psychological components of performance and recovery,[1,2] these parameters could only be controlled to a certain extent. While objective measures were assumed to be maximized at baseline, this may have not been the case considering PRS at baseline ranged from 6 to 10. Second, the subjective nature of perceptual scales and their relationship to physiological outcomes is understood to exhibit interindividual variability and should be considered on an individual basis rather than applied to the entire group.[25,26] The repeated measure correlations combined with the frequency table (Table 2) appear to validate this notion. So, while a PRS score of 6 may mean a different level of performance recovery between 2 individuals, the relationship between PRS and objective recovery of performance markers appear to be stable within an individual.

Though this study was designed to assess the applicability of PRS rather than the physiological mechanisms that mediate it, some may be proposed based upon what is understood about other subjective scales used in performance assessment. For example, ratings of perceived exertion (RPE) is a psychophysiological scale that allows an individual to subjectively estimate their effort during exercise. As such, RPE is well correlated with a variety of peripheral measures including heart rate, pain, and ventilation,[27] as well as psychological factors.[28] More recently, RPE was also shown to correlate with increased activity in areas specific to motor control.[29] While debate exists whether RPE is derived from afferent feedback, central motor command, or a combination, the influence of afferent feedback is stressed in all models.[30] Indeed, RPE has been hypothesized to be both the conscious manifestations of physiological feedback and psychological sensations, as well as a primary regulator of exercise performance.[31,32] It may be that PRS acts similarly, but as an *estimation of future* performance based on the current physiological and psychological state rather than the *manifestation of current* physiological and psychological distress due to acute exertion.

Practical Applications

The PRS appears to be a valuable addition to a battery of monitoring tools that practitioners may use as it is an accurate cost- and time-effective method to monitor athletes' daily recovery status. Practitioners should assess PRS after an athlete has had the opportunity to warm-up.[3] Should the athlete report an extremely high or low PRS score (this score will be dependent upon the individual) then immediate alterations to their program can be made by augmenting intensity or volume to take advantage of this reported estimation of perceived recovery or by altering training in a way to achieve the load desired. In addition, PRS scores can be recorded and monitored daily to assess trends in estimated recovery status and compare them to the practitioner's expectations and prior training volume. A trend of PRS scores lower than expected by the practitioner may indicate that the athlete is not handling the load and modifications to their training and/or recovery program need to occur before nonfunctional overreaching and overtraining occur.

While findings from the current study are promising, the generalizability of our findings is limited in scope, and additional research on the widespread utility of PRS as a subjective measure of recovery status is warranted. Specifically, future research should examine how PRS estimations are derived and the physiological (eg, hydration, sleep, environmental factors) and psychological (eg, stress, anxiety, and self-efficacy) factors that modulate subjective recovery. Practitioners and researchers should exercise caution when using PRS in practice, incorporating it as a supplement to other measurements of recovery and readiness to perform. The current study assessed the accuracy of PRS following a fatiguing high-volume back squat protocol indicative of a hypertrophy type of resistance exercise workout. Additional research is required to assess the accuracy of PRS following strength and power-based (ie, high intensity, low volume) routines and the utility of PRS throughout the course of periodized training regimens employing multiple exercises and muscle groups. Nevertheless, findings from the current study are promising.

Conclusion

The present study is the first to quantify the relationship between PRS and recovery of performance following a bout of high-volume back squat protocol. Strong correlations were observed between PRS and performance metrics when accounting for between-subject variability. These results support the use of PRS as a daily metric of recovery following resistance exercise in resistance trained males. Nonetheless, the generalizability of our findings is limited in scope, and additional research on the widespread utility of PRS as a subjective measurement of recovery status is warranted.

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